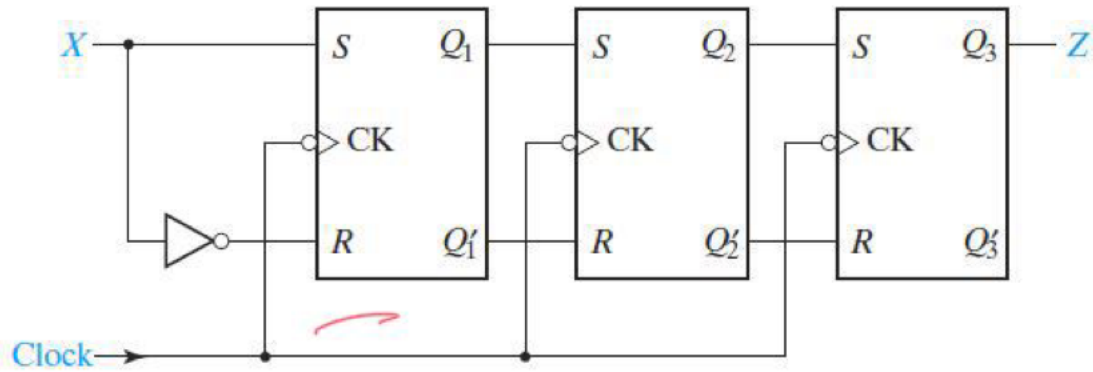


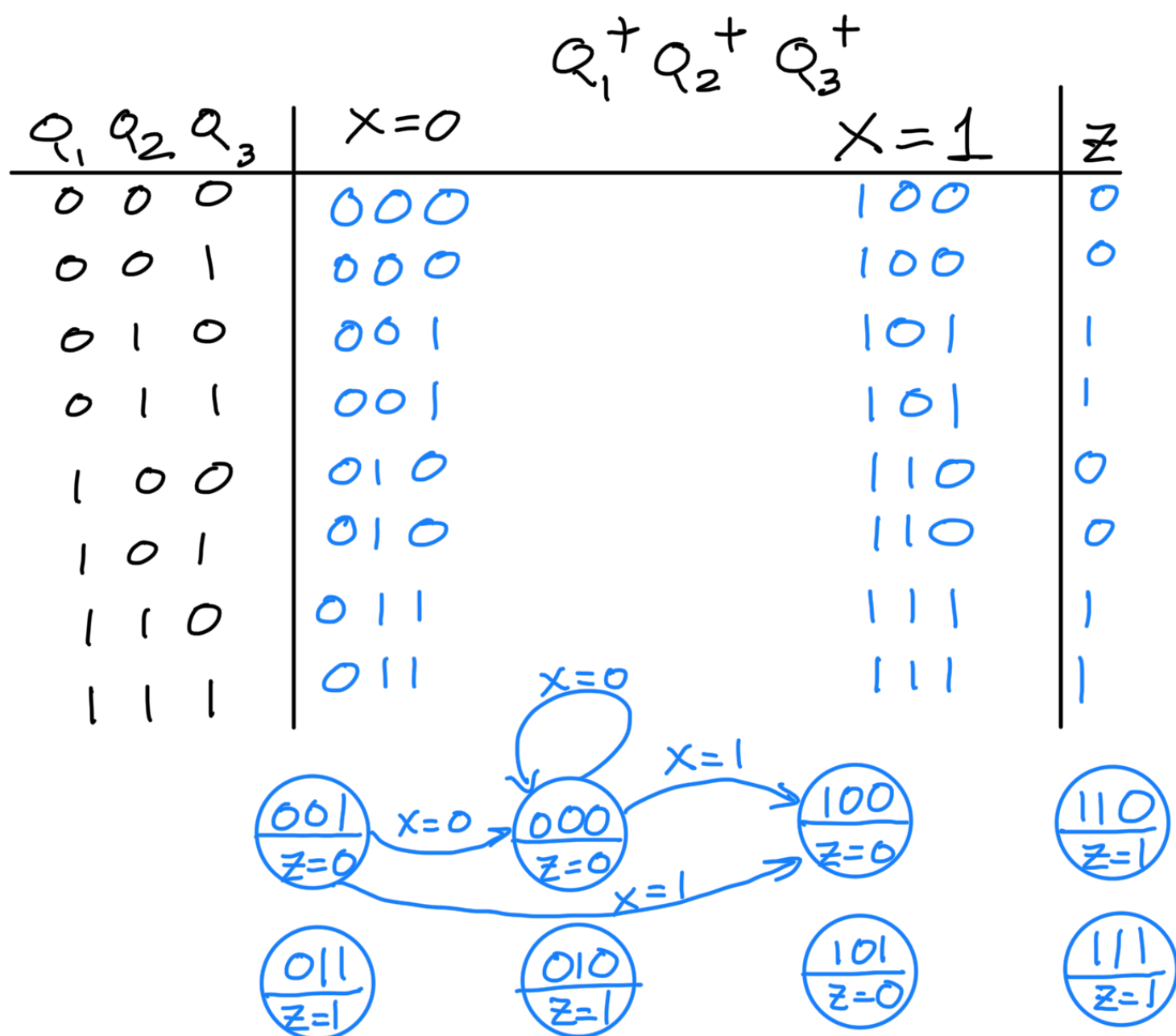
Unit 13 Solutions

13.2 Construct a transition graph for the shift register shown. (X is the input, and Z is the output.) Is this a Mealy or Moore machine?

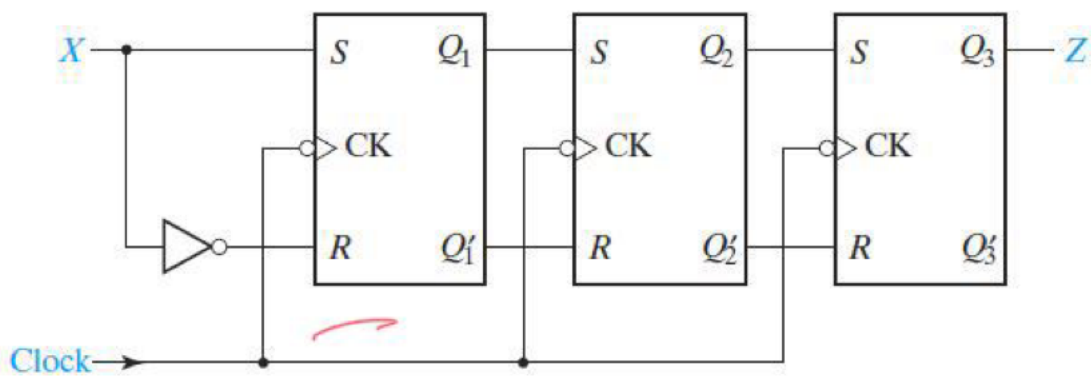


X is not specified at start. So, we will have to look at all combinations of inputs, states. X passes through a D before reaching R input to S-R FF, so $S \neq R$ and therefore, $S=R=1$ will never occur. Also, Q and Q' outputs become inputs to S and R respectively of next FF in sequence. So, once again, $S=R=1$ will never occur for remaining FFs either.

$Z = Q_3^+$. $\therefore$ This is a Moore machine
 This is a serial shift register. So the state of exactly one FF can change on any single clock cycle.



13.2 Construct a transition graph for the shift register shown. (X is the input, and Z is the output.) Is this a Mealy or Moore machine?



- 13.5 Below is a transition table with the outputs missing. The output should be $Z = X'B' + XB$.
- Is this a Mealy machine or Moore machine?
 - Fill in the outputs on the transition table.
 - Give the transition graph.
 - For an input sequence of $X = 10101$, give a timing diagram for the clock, X , A , B , C , and Z . State changes occur on the rising clock edge. What is the correct output sequence for Z ? Change X between rising and falling clock edges so that we can see false outputs, and indicate any false outputs on the diagram.

ABC	$A+B+C$	
	$X = 0$	$X = 1$
000	011	010
001	000	100
010	100	100
011	010	000
100	100	001

a) Mealy machine $\therefore Z = \underline{X}'B' + \underline{X}B$ depends on input (and FF states).

b)

	$A^+ B^+ C^+$		Z	
ABC	X=0	X=1	X=0	X=1
0 <u>0</u> 0	011	010	$B=0$ 1	0
0 <u>0</u> 1	000	100	$B=0$ 1	0
0 <u>1</u> 0	100	100	$B=1$ 0	1
0 <u>1</u> 1	010	000	$B=1$ 0	1
1 <u>0</u> 0	100	001	$B=0$ 1	0

c)

